SOFT COMPUTING (CSC425) DIMENSIONALITY REDUCTION AND CLASSIFICATION

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Linear Discriminate Analysis (LDA)

LINEAR DISCRIMINANT ANALYSIS (LDA)

LDA is a supervised classification method originally developed in 1936 by R.A. Fisher.

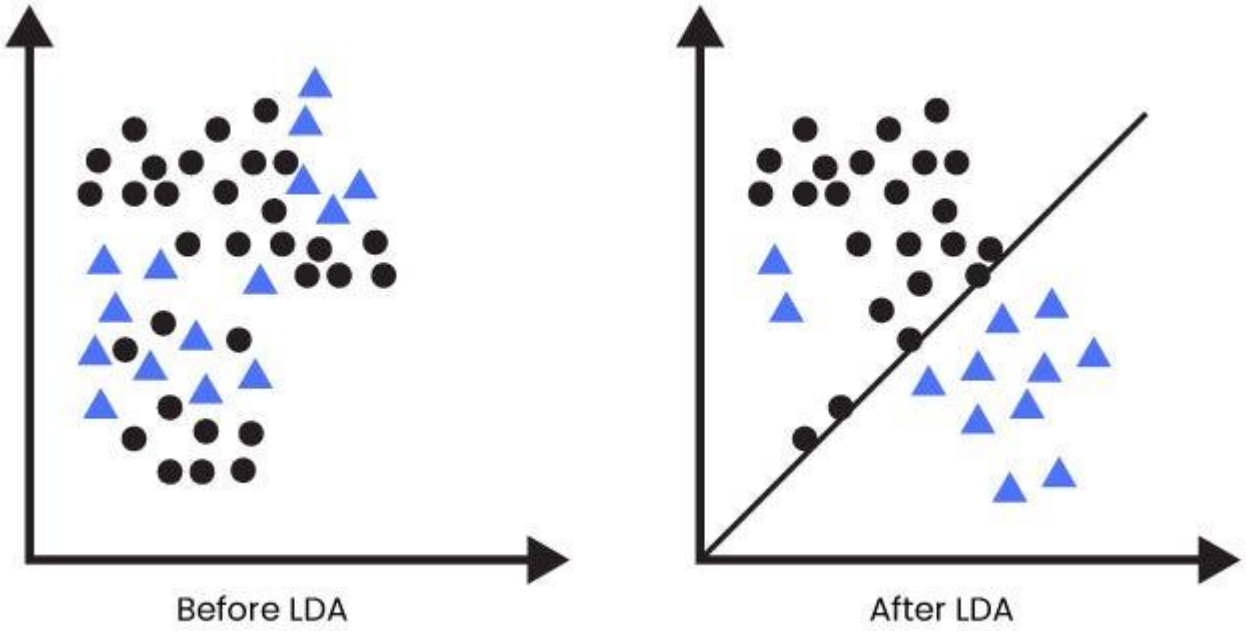
LDA is a dimensionality reduction and classification technique commonly used in machine learning and pattern recognition.

It is based upon the concept of searching for a linear combination of the variables that best separates two classes.

It is simple, mathematically robust and often produces models whose accuracy is as good as more complex methods.

The main purpose of LDA is to find the line (or plane) that best separates data points belonging to different classes.

LINEAR DISCRIMINANT ANALYSIS (LDA)



LDA ALGORITHM

- LDA is based upon the concept of searching for a linear combination of variables (predictors) that best separates two classes (targets).
- The key idea behind LDA is that the decision boundary should be chosen such that it **maximizes** the distance between the means of the two classes while simultaneously minimizing the variance within each classes data or within-class scatter.
- LDA works by projecting the data onto a lower-dimensional space while preserving as much of the class information as possible and then finding the hyperplane that maximizes the separation between the classes based on Fisher criterian
- To capture the notion of separability, Fisher defined the following score function:

FISHER'S LDA – MODEL

$$Z = \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_d x_d$$

$$S(\beta) = \frac{\beta^T \mu_1 - \beta^T \mu_2}{\beta^T C \beta}$$

Score function

$$\beta = C^{-1}(\mu_1 - \mu_2)$$

Model coefficients

Given the score function, the problem is to estimate the linear coefficients that maximize the score.

$$\beta = C^{-1}(\mu_1 - \mu_2)$$ *Model coefficients*

$$C = \frac{1}{n_1 + n_2}(n_1 C_1 + n_2 C_2)$$ *Pooled covariance matrix*

Where:

β : *Linear model coefficients*

C_1, C_2 : *Covariance matrices*

μ_1, μ_2 : *Mean vectors*

LDA – TRAINING

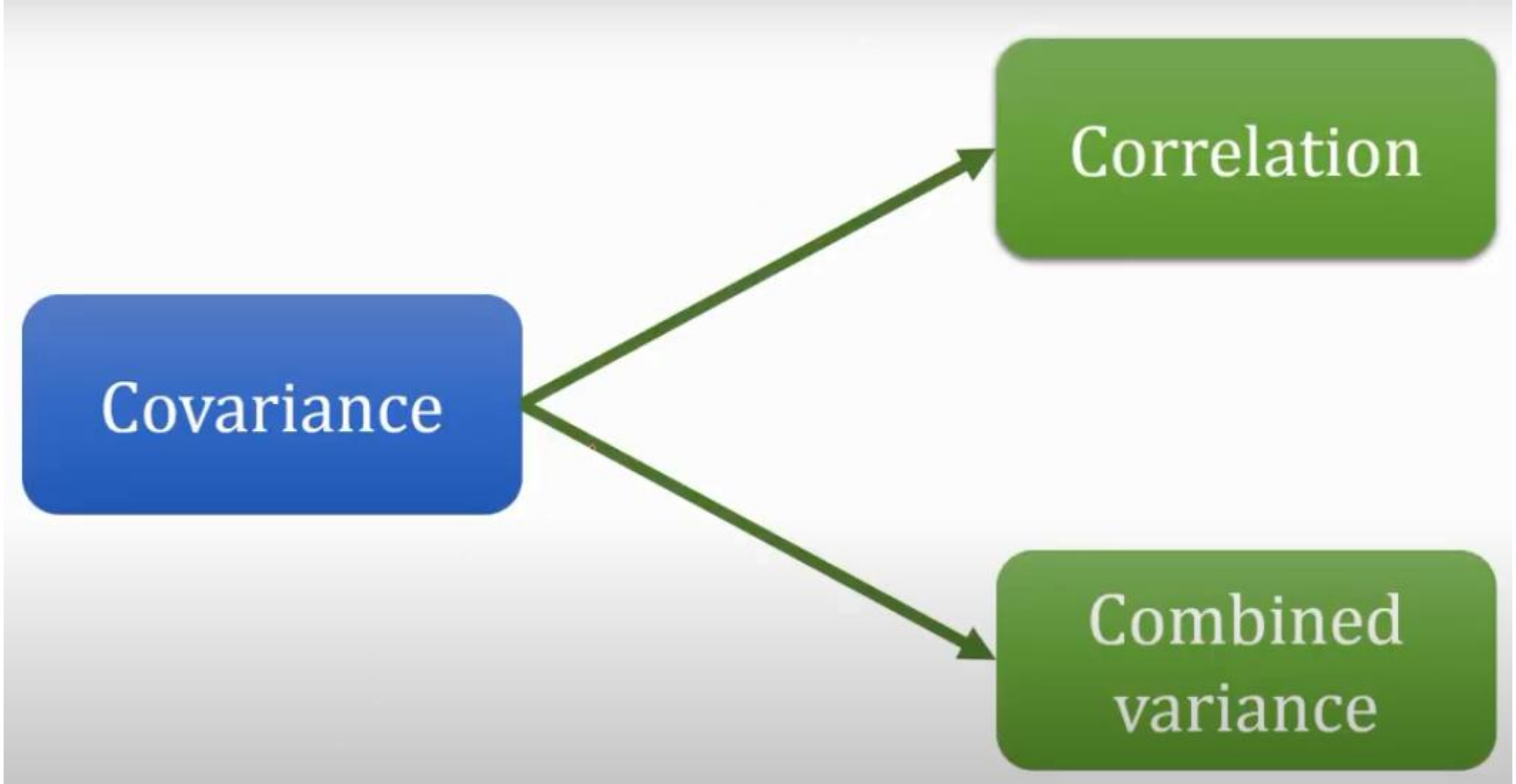
Mean vector

$$\mu = (\mu_1, \mu_2, \mu_3, \ldots, \mu_d) \quad \Rightarrow \quad \mu = \frac{\sum x}{n}$$

Covariance matrix

$$C = \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1d} \\ C_{21} & C_{22} & \cdots & C_{2d} \\ \cdots & \cdots & \cdots & \cdots \\ C_{d1} & C_{d2} & \cdots & C_{dd} \end{bmatrix} \quad \Rightarrow \quad C = \frac{\sum (x - \mu_x)(y - \mu_y)}{n}$$

COVARIANCE

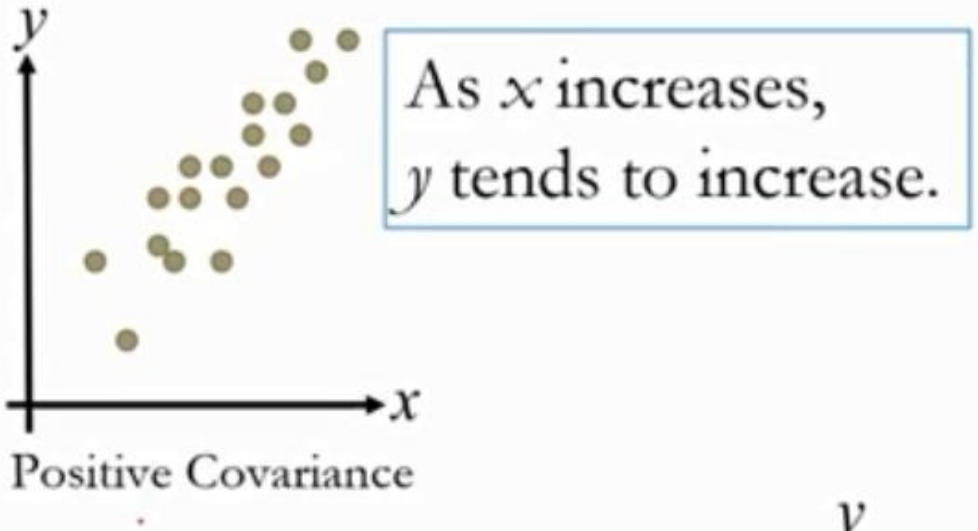


COVARIANCE

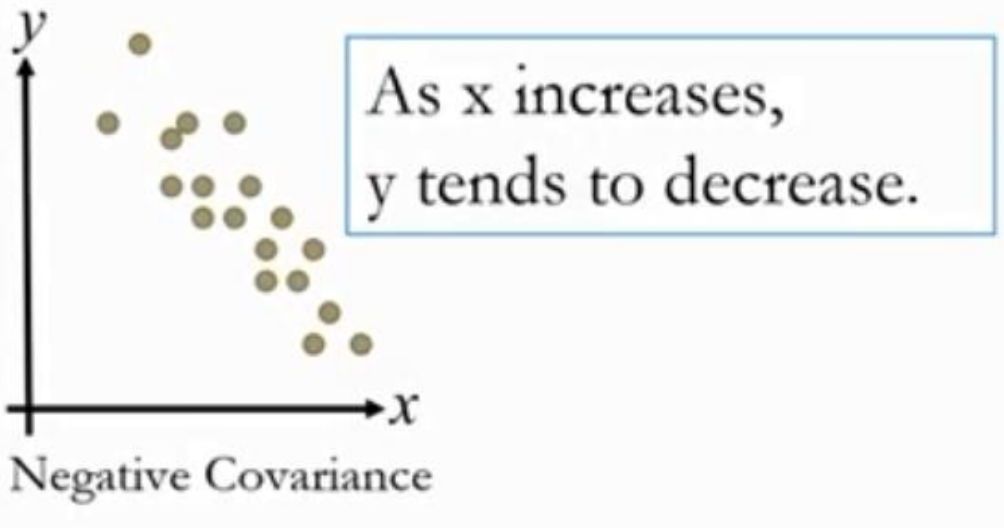
- **Covariance** is a measure of the relationship between two random variables.

- **Positive covariance:** Indicates that two variables tend to move in the same direction.
- **Negative covariance:** Reveals that two variables tend to move in inverse directions.

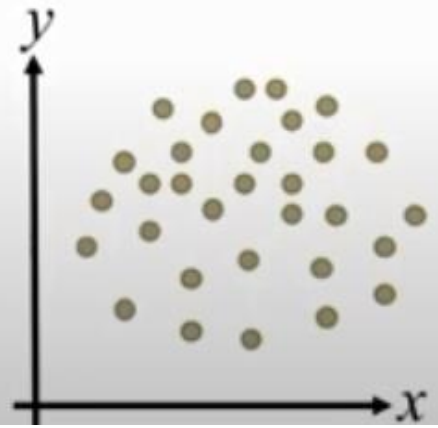
COVARIANCE



Positive Covariance



Negative Covariance



LDA – MODEL ASSESSMENT

One way of assessing the effectiveness of the discrimination is to calculate the **Mahalanobis distance** between two groups.

A distance greater than 3 means that in two averages differ by more than 3 standard deviations.

It means that the overlap (probability of misclassification) is quite small.

$$\Delta^2 = \beta^T(\mu_1 - \mu_2)$$

Δ : *Mahalanobis distance between two groups*

LDA – PREDICTION

A new point is classified by projecting it onto the maximally separating direction and classifying it as **Class1 if:**

$$\beta^T\left(x - \left(\frac{\mu_1 + \mu_2}{2}\right)\right) > \log\frac{p(c_1)}{p(c_2)}$$

Coefficients vector

Data vector

Mean vector

Class probability

LDA -EXAMPLE

Suppose we received a dataset from a bank regarding its small business clients who defaulted (red square) and those that did not (blue circle) separated by delinquent days (DAYSDELQ) and number of months in business (BUSAGE).

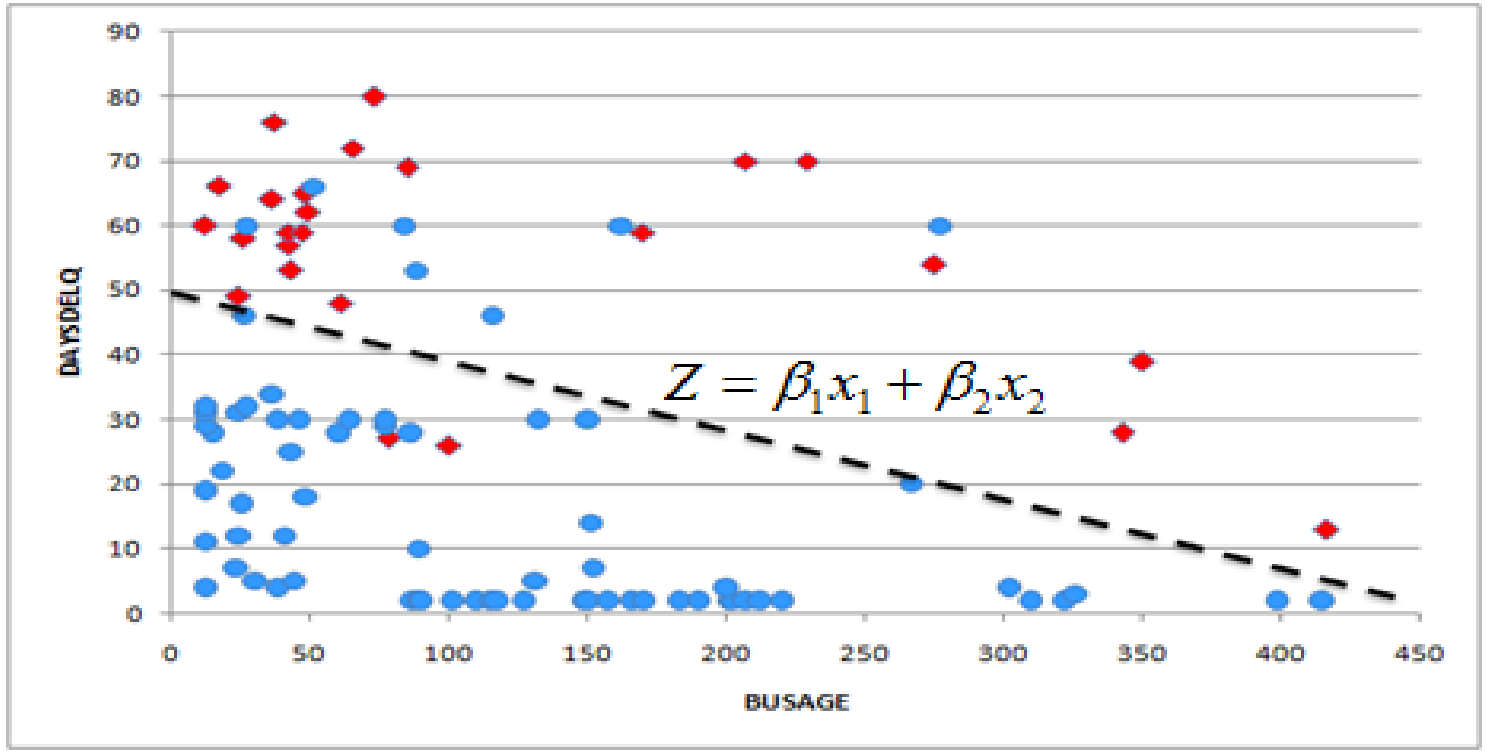
Use LDA to find an optimal linear model that best separates two classes (default and non-default).

DATA SET

	BUSAGE	DAYSDELQ	DEFAULT
1	87	2	N
2	89	2	N
3	90	2	N
4	90	2	N
5	101	2	N
6	110	2	N
7	115	2	N
8	115	2	N
9	115	2	N
10	117	2	N
11	127	2	N
12	149	2	N
13	150	2	N
14	157	2	N
15	166	2	N
16	170	2	N
17	183	2	N
18	183	2	N
19	190	2	N
20	201	2	N
21	202	2	N

	BUSAGE	DAYSDELQ	DEFAULT
39	152	7	N
40	89	10	N
41	24	12	N
42	41	12	N
43	151	14	N
44	25	17	N
45	48	18	N
46	12	19	N
47	267	20	N
48	18	22	N
49	43	25	N
50	15	28	N
51	60	28	N
52	86	28	N
53	12	29	N
54	77	29	N
55	38	30	N
56	46	30	N
57	64	30	N
58	77	30	N
59	132	30	N

	BUSAGE	DAYSDELQ	DEFAULT
1	100	26	Y
2	78	27	Y
3	61	48	Y
4	343	28	Y
5	24	49	Y
6	43	53	Y
7	417	13	Y
8	275	54	Y
9	42	57	Y
10	26	58	Y
11	42	59	Y
12	47	59	Y
13	170	59	Y
14	12	60	Y
15	49	62	Y
16	36	64	Y
17	48	65	Y
18	17	66	Y
19	85	69	Y
20	350	39	Y
21	207	70	Y


90
80
70
60
50
40
30
20
10
0
DAYSDELQ
0
50
100
150
200
250
300
350
400
450
BUSAGE
$Z = \beta_1 x_1 + \beta_2 x_2$

LDA TRAIN - EXAMPLE

The first step is to calculate the mean (average) vectors, covariance matrices and class probabilities.

Class (DEFAULT)	Count	Probability	Statistics	BUSAGE	DELQDAYS
N	n_1=75	$p(c_1)$=0.75	Means Vector (μ)	116.23	16.89
			Covariance Matrix ($\mathbf{C}$)	$\begin{bmatrix} 9323 & -607 \\ -607 & 333 \end{bmatrix}$	
Y	n_2=25	$p(c_2)$=0.25	Means Vector (μ)	115.04	55.32
			Covariance Matrix ($\mathbf{C}$)	$\begin{bmatrix} 14009 & -1053 \\ -1053 & 287 \end{bmatrix}$	

LDA TRAINING

Then, we calculate pooled covariance matrix and finally the coefficients of the linear model.

$$C = \frac{1}{n_1 + n_2}(n_1 C_1 + n_2 C_2) = \begin{bmatrix} 10495 & -718 \\ -718 & 322 \end{bmatrix}$$

$$\beta = C^{-1}(\mu_1 - \mu_2) = \begin{bmatrix} -0.0095 & -0.1408 \end{bmatrix}$$

$$Z = -0.0095\ BUSAGE - 0.1408\ DAYSDELQ$$

LDA TRAINING

A *Mahalanobis distance* of 2.32 shows a small overlap between two groups which means a good separation between classes by the linear model.

$$\Delta^2 = \beta^T(\mu_1 - \mu_2) = 5.40$$
$$\Delta = 2.32$$

LDA PREDICTION

Assume we have a point with BUSAGE=111 and DAYSDELQ=24.

X=[111 24]

$$\beta^T\left(x - \left(\frac{\mu_1 + \mu_2}{2}\right)\right) > \log\frac{p(c_1)}{p(c_2)}$$

Coefficients vector

Data vector

Mean vector

Class probability

$$\begin{bmatrix} -0.0095 \\ -0.1408 \end{bmatrix} \left(\begin{bmatrix} 111 & 24 \end{bmatrix} \left(\frac{\begin{bmatrix} 116.23 & 16.89 \end{bmatrix} + \begin{bmatrix} 115.04 & 55.32 \end{bmatrix}}{2} \right) \right)$$

$$\overset{?}{>} \log \frac{0.75}{0.25}$$

LDA-PREDICTION

$$\beta^T\left(x-\left(\frac{\mu_1+\mu_2}{2}\right)\right) > -\log\frac{p(c_1)}{p(c_2)} = -\log\left(\frac{0.75}{0.25}\right) = -1.1$$

BUSAGE	DAYSDELQ	DEFAULT	Z Score	$\beta^T\left(x-\left(\frac{\mu_1+\mu_2}{2}\right)\right)$	Prediction
87	2	N	-1.1081	5.0752	N
89	2	N	-1.1271	5.0562	N
90	2	N	-1.1366	5.0466	N
116	46	N	-7.5788	-1.3965	Y
88	53	N	-8.2984	-2.1155	Y
42	57	Y	-8.4246	-2.2405	Y
26	58	Y	-8.4134	-2.2289	Y
42	59	Y	-8.7062	-2.5221	Y
343	28	Y	-7.2009	-1.0245	N
24	49	Y	-7.1272	-0.9426	N

PREDICTORS CONTRIBUTION

A simple linear correlation between the model scores and predictors can be used to test which predictors contribute significantly to the discriminant function.

Correlation varies from -1 to 1, with -1 and 1 meaning the highest contribution but in different directions and 0 means no contribution at all.

EXTENSIONS TO LDA

Quadratic Discriminant Analysis (QDA): Each class uses its own estimate of variance (or covariance when there are multiple input variables).

QUADRATIC DISCRIMINANT ANALYSIS (QDA)

QDA is a general discriminant function with a quadratic decision boundaries which can be used to classify datasets with two or more classes.

QDA has more predictability power than LDA but it needs to estimate the covariance matrix for each classes.

QDA enables it to create curved, non-linear decision boundaries.

LDA APPLICATIONS

1. Face Recognition: LDA is frequently employed in face recognition systems. By reducing the dimensionality of face images while preserving the essential information for distinguishing between individuals.

2. Medical Diagnosis: In medical fields, LDA can aid in diagnosing diseases or conditions based on patient data. For instance, it can be used to classify patients as healthy or suffering from a particular disease based on a set of medical features.

3. **Biometrics:** LDA can also be applied to other biometric identification systems, such as fingerprint recognition and iris recognition.

4. Quality Control and Manufacturing: LDA can assist in identifying defects in products by classifying items as defective or non-defective based on various measurements or attributes.

5. Document Classification: LDA can be used for categorizing documents into different classes or topics.

LDA PROS

- **Dimensionality Reduction with Class Separation:** LDA aims to maximize the separation between classes while reducing the dimensionality of the data.

- **Works Well for Small Sample Sizes**: LDA can handle situations where the number of samples is small compared to the number of features.

- **Interpretable Results:** The reduced-dimensional representation obtained through LDA can often be more interpretable than the original feature space. This can aid in understanding the important factors driving the classification.

- **Data Visualization**: The reduced-dimensional space generated by LDA can be visualized, making it easier to observe the separation between classes and the distribution of data points.

- **Robust to Outliers**: LDA is less sensitive to outliers compared to other methods like k-nearest neighbors. This is due to its reliance on class means and variances rather than individual data points.

LDA CONS

Sensitive to Class Distribution: LDA assumes that the classes have approximately equal covariance matrices and follow a Gaussian distribution. If these assumptions are not met, the performance of LDA can degrade. In cases where the assumptions don't hold, techniques like Quadratic Discriminant Analysis (QDA) or non-parametric methods might be more appropriate.

Doesn't Handle Nonlinear Relationships: LDA assumes linear relationships between features and classes. If the relationships are nonlinear, LDA might not capture the underlying patterns accurately.

SUMMARY

- Linear Discriminant Analysis (LDA) is a classification method for separating two classes.
- It is simple, mathematically robust and often produces models whose accuracy is as good as more complex modern methods.
- LDA is a powerful tool for dimensionality reduction.
- LDA is not susceptible to the “curse of dimensionality” like many other machine learning algorithms and work well with limited sample size of data.
- LDA is particularly well-suited for tasks such as facial recognition where data from different sources needs to be compared.
- LDA can be used for a variety of tasks such as classification, dimensionality reduction, and feature selection.

LDA Applications

https://medium.com/aimonks/linear-discriminant-analysis-lda-in-machine-learning-example-concept-and-applications-37f27e7c7e98

More Examples on LDA

https://people.revoledu.com/kardi/tutorial/LDA/Numerical%20Example.html

Questions?